

Design of Dual-Band T-Shaped, U-Shaped and H-Shaped Microstrip Antennas for Wireless Applications

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Abstract— This work presents a study comparing Dual-band T-shaped, U-shaped, and H-shaped microstrip patch antennas designed to operate at a frequency of 5.25 GHz for modern wireless communication systems. All antenna designs are built on the same FR-4 (epoxy material) substrate, which ensures consistent design parameters and allows for an objective comparison of their performance. To improve radiation performance, design-specific enhancement techniques are used. The T-shaped antenna uses a defected ground structure (DGS) to achieve a wider bandwidth, while the U-shaped and H-shaped antennas include coplanar parasitic elements to increase gain through effective near-field electromagnetic coupling. The proposed antennas are designed, simulated, and optimized using ANSYS HFSS, with performance evaluated in terms of reflection coefficient, bandwidth, gain, and radiation patterns. Simulation results show better impedance matching, gain, and stable radiation patterns within the intended operating bands. Because of their compact size, low-profile design, and cost-effective fabrication, these antenna designs are well-suited for wireless communication applications.

Key words— Microstrip patch antenna (MPA), dual-band, DGS, WLAN (Wireless Local Area Network), WiMAX (Worldwide Interoperability for Microwave Access), coplanar parasitic elements, Vector Network Analyzer (VNA).

I. INTRODUCTION

The rapid growth in communication technologies has strengthened the need for small, low-profile, and high-performance antennas that can support multiband operations. Applications such as WLAN (Wireless Local Area Networks), WiMAX (Worldwide Interoperability for Microwave Access), and other latest wireless systems that operates below 6 GHz frequency range require antennas that offer sufficient impedance bandwidth, better gain, and stable radiation patterns while minimizing size and complexity in production. In this situation, MPA's are still favored due to their flat design, light weight, low cost, and compatibility with PCB technology.

However, traditional rectangular microstrip patch antennas struggle with limited impedance bandwidth and low realized gain. These problems come from the high-quality factor associated with the resonant cavity which is due to the radiating patch and the ground plane, along with dielectric and

surface-wave losses, especially when using low-cost substrates like FR-4. which results, substantial exploration has concentrated on finding ways to master these restrictions through structural changes and performance improvement techniques.

Among the reported methods, slot-based patch geometry modification, DGS, and parasitic element methods will be better for increasing bandwidth and gain. Slot-loaded patch designs like T-shaped, U-shaped, and H-shaped configurations create additional surface current paths, allowing for multi-resonant behavior and compact dual-band operation. At the same time, DGS helps to enhance impedance characteristics, while parasitic elements increase radiation efficiency through electromagnetic coupling.

This research shows the relative investigation of dual-band T-shaped, U-shaped, and H-shaped MPA's operating at 5.25 GHz. The T-shaped antenna uses a DGS method to enhance the bandwidth and performance, while the U-shaped and H-shaped antennas employ coplanar parasitic elements to increase gain. All antenna configurations are built on a common FR-4 substrate to maintain consistency and allow for a fair comparison. The antennas are modeled and optimized using ANSYS HFSS, and their performance is assessed in terms of return loss, Impedance bandwidth, Gain, and radiation characteristics. These designs aim to serve WLAN, WiMAX, and general wireless communication applications.

II. LITERATURE REVIEW

The rectangular MPA's are commonly utilized in recent wireless communication systems because of their ease of manufacturing and similarity with planar circuits [1]–[5]. However, conventional MPA's have narrow impedance bandwidth and limited gain, which can restrict their characteristics in broadband and multiband wireless applications [1], [3]. To overcome these conflicts, they have focused on improving antenna performance by modifying their geometry and using various techniques.

One effective method involves Defected Ground Structures (DGS). DGS can increase bandwidth and radiation efficiency by minimizing surface-wave propagation. The basic concept of DGS was introduced by Ahn et al. [6]. Multiple works displayed that applying DGS to microstrip

antennas results in improved bandwidth and impedance characteristics [8], [14], [17]. Changes to the ground material have also been shown to enhance multiband performance and radiation stability without significantly increasing the length and width of the antenna [21].

Another approach to boost performance is by using parasitic elements, which improve radiation characteristics through electromagnetic coupling. Yang and Rahmat-Samii [7] showed how parasitic structures can increase antenna efficiency. Additionally, Sung [10] and Bhattacharya and Gupta [19] indicated gains and bandwidth improvements with parasitic patches. Slot-loaded and shaped patch antennas can achieve multiband behavior by activating various resonant modes [12], [13], [18].

U-shaped and H-shaped slot antennas have been rigorously explored for compact dual-band and multiband applications. U-shaped antennas provide wide impedance bandwidth and stable radiation patterns, especially for WLAN systems [9], [24]. H-shaped configurations offer better impedance control and symmetry for multiband operation [23]. Several miniature multiband antennas have been established for WLAN and WiMAX frequency bands, highlighting the requirement for improved bandwidth and stable radiation performance [15], [16], [22].

The literature review shows that DGS techniques effectively achieve wide bandwidth, while strategies using parasitic elements allow for controlled gain enhancement and stable multiband performance. However, there is limited comparative analysis of T-shaped antennas with DGS and U- and H-shaped antennas with coplanar parasitic elements on a common substrate. This gap motivates the current study.

III. METHODOLOGY

A. Antenna Design Steps

Step 1: Patch Width (W)

The width of the rectangular patch is given by:

$$W = \frac{c}{2f_0 \sqrt{\frac{\epsilon r + 1}{2}}} \quad (1)$$

where c represents the velocity of light in free space.

Step 2: The Effective Dielectric Constant (ϵ_{eff})

The ϵ_{eff} is computed as:

$$\varepsilon_{eff} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2} \left(1 + 12 \frac{h}{w} \right)^{-1/2} \quad (2)$$

This accounts for the fringing fields at the patch periphery.

Step 3: Effective Length (L_{eff})

he patch effective length is given by:

$$L_{eff} = \frac{c}{2f_0 \sqrt{\epsilon_{r_{eff}}}} \quad (3)$$

Step 4: Length Extension (ΔL)

The extension in patch length due to fringing fields is:

$$\Delta L = 0.412h \frac{(\varepsilon_{eff}+0.3)\left(\frac{w}{h}+0.264\right)}{(\varepsilon_{eff}-0.258)\left(\frac{w}{h}+0.8\right)} \quad (4)$$

Step 5: Length of Patch (L)

The actual patch length is obtained as:

$$L = L_{eff} - 2\Delta L \quad (5)$$

Step 6: Substrate Dimensions (Ls & Ws)

To ensure proper isolation and minimal fringing, the substrate dimensions are taken as:

$$L_s = L + 2 * 6h \quad (6)$$

$$W = W + 2 * 6h \quad (7)$$

The calculated values formed the baseline geometries for the designed antenna a prior to further optimization, which is described in the subsequent sections.

B. Design Methodology and Patch Geometry

This Paper configurations consist of T-shaped, U-shaped, and H-shaped MPA's, all designed on an FR-4 substrate with relative permittivity $\epsilon_r = 4.4$ and loss tangent, $\delta = 0.02$. A uniform substrate thickness is selected to balance impedance bandwidth enhancement and surface-wave suppression while maintaining ease of fabrication. A continuous ground plane is used for the U-shaped and H-shaped antennas, whereas a DGS technique is introduced for the T-shaped antenna.

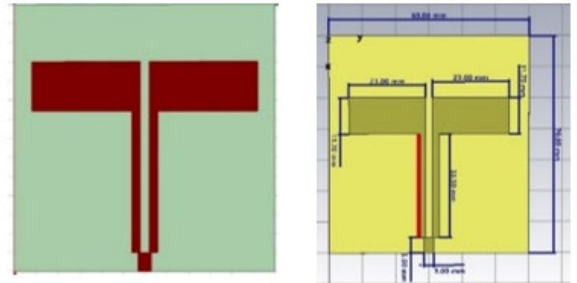


Fig (a)

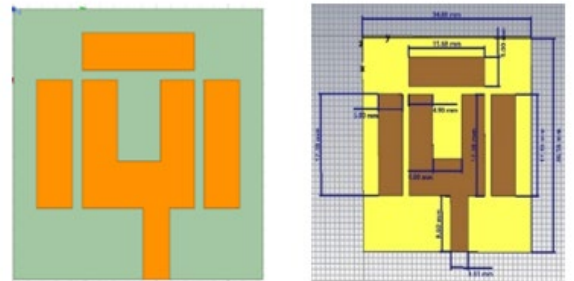


Fig (b)

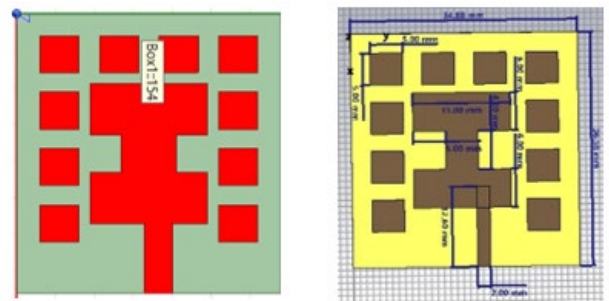


Fig (c)

Fig. 1. Top view of the Proposed Antennas (a) T-Shaped patch fed by a 50- Ω microstrip line. (b) U-Shaped Patch with Coplanar Patch Elements. (c) H-Shaped Patch with Coplanar Patch Elements.

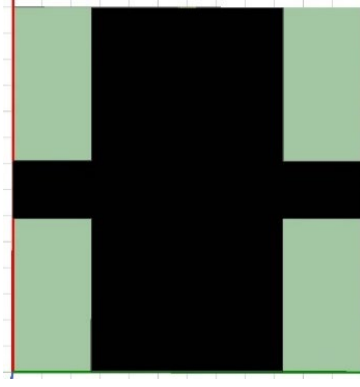


Fig. 2. Bottom view showing the modified + shaped defected ground structure (DGS).

IV. RESULTS AND DISCUSSION

A. Return Loss and Bandwidth Analysis

T-Shaped Microstrip Patch Antenna with DGS: The simulated return loss response of the T-shaped MPA incorporating a DGS, as shown in Fig. 3, confirms dual-band operation with resonances at 3.4 GHz and 5.41 GHz. The lower resonant band exhibits a narrow impedance bandwidth of 20 MHz, which results from the high-Q resonance introduced by the combined effect of the T-shaped geometry and ground-plane defect. In contrast, the upper band demonstrates a substantially enhanced bandwidth of 1.052 GHz, indicating effective suppression of surface-wave losses and improved impedance matching because of the DGS. The wide upper-band bandwidth highlights the effectiveness of the defected ground structure in achieving broadband performance around the 5.25 GHz WLAN band.

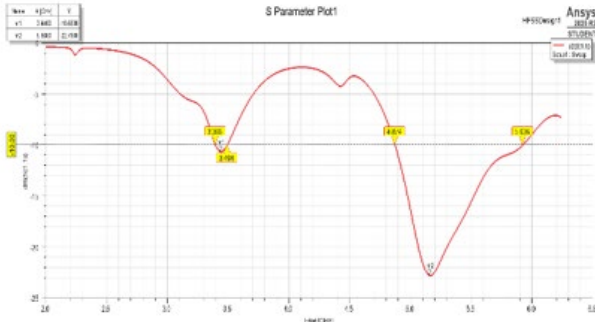


Fig (a)

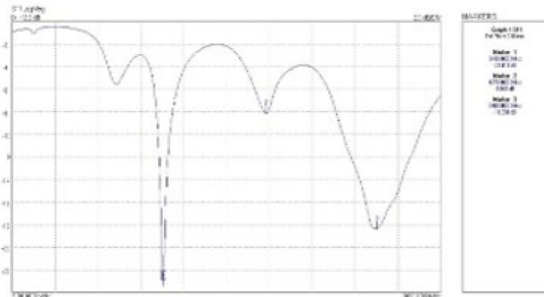


Fig (b)

Fig. 3. (a) Simulated and (b) VNA Return loss (S_{11}), Results of T-shape Antenna.

U-Shaped MPA with Coplanar Parasitic Elements:

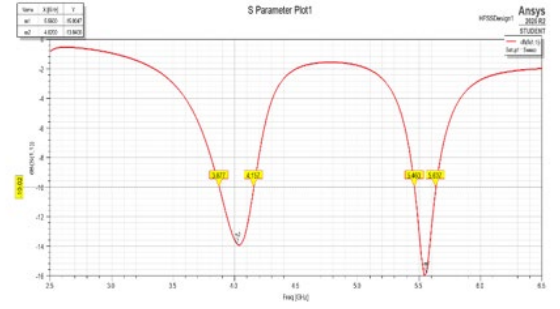


Fig (a)

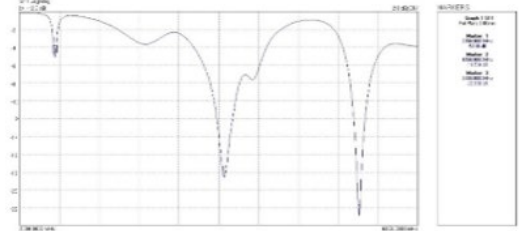


Fig (b)

Fig. 4. (a) Simulated and (b) VNA Return loss (S_{11}), Results of U-shape Antenna.

Fig. 4 illustrates the S_{11} characteristics of the U-shaped microstrip patch antenna employing a coplanar parasitic patch element. The antenna exhibits dual-band behavior with a lower resonance centered at 4.0 GHz and 280 MHz bandwidth, originating through electromagnetic coupling between the U-slot-induced mode and the fundamental patch mode. The upper resonant band, centered at 5.41 GHz, achieves a bandwidth of 174 MHz, which is enhanced by near-field electromagnetic coupling with the coplanar parasitic element. The results demonstrate that the U-shaped configuration provides a balanced bandwidth distribution across both operating bands, making it suitable for multiband wireless applications.

H-Shaped MPA with Coplanar Parasitic Element:

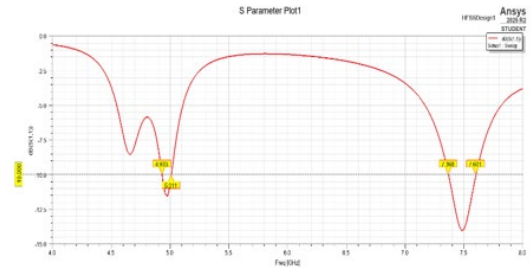


Fig (a)

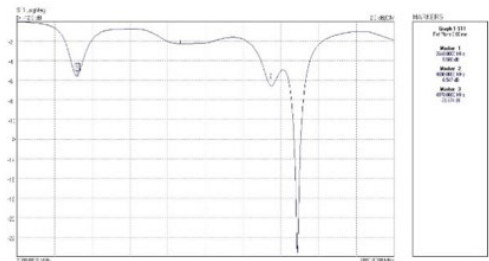


Fig (b)

Fig. 5. (a) Simulated and (b) VNA Return loss (S_{11}), Results of H-shape Antenna

The return loss behavior of the H-shaped MPA integrated with coplanar parasitic elements, depicted in Fig. X(c), reveals stable dual-band operation with resonant frequencies at 4.9 GHz and 5.41 GHz. The lower band achieves an impedance bandwidth of 78 MHz, while the upper band exhibits a comprehensive bandwidth of 230 MHz. The symmetrical H-shaped slot geometry ensures balanced surface current distribution and stable impedance characteristics, whereas the coplanar parasitic element effectively enhances the upper-band bandwidth by increasing the effective radiating aperture. This configuration demonstrates consistent impedance performance and controlled bandwidth enhancement suitable for wireless communication technologies.

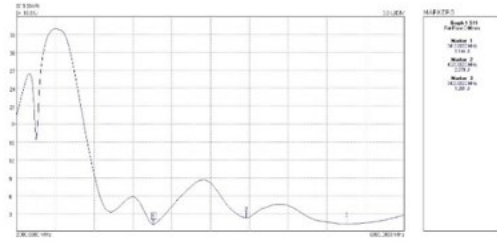
B. Gain Analysis

The realized gain of the proposed antennas was evaluated at the upper operating frequency of 5.41 GHz, where all configurations exhibit good impedance matching. The T-shaped antenna with DGS achieves the highest gain of 6.15 dBi, attributed to effective surface-wave suppression and enhanced radiation efficiency. The U-shaped antenna employing a coplanar parasitic element exhibits a gain of 5.2 dBi, resulting from near-field electromagnetic coupling that increases the effective radiating aperture. The H-shaped antenna with coplanar parasitic coupling attains a slightly higher gain of 5.40 dBi, due to its symmetrical geometry and balanced current distribution. Overall, the gain results are consistent with the impedance and bandwidth characteristics, confirming that DGS-based enhancement provides maximum gain, while parasitic-coupled U- and H-shaped antennas offer stable and efficient radiation performance suitable for WLAN, WiMAX, and wireless communication applications.

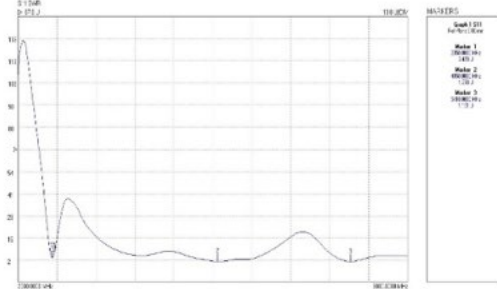
C. VSWR plots

The VSWR performance of the proposed antennas was evaluated at the upper operating frequency of 5.41 GHz. The T-shaped antenna with DGS exhibits a VSWR of 1.281, indicating efficient impedance matching at the feed point. The U-shaped antenna with a coplanar parasitic element achieves an improved VSWR of 1.151, attributed to effective near-field electromagnetic coupling, while the H-shaped antenna demonstrates the lowest VSWR of 1.13 due to its symmetrical geometry and balanced current distribution. In all cases, the VSWR values remain well below the acceptable limit of 2, confirming better impedance matching and validating the return loss performance for WLAN, WiMAX, and wireless communication applications.

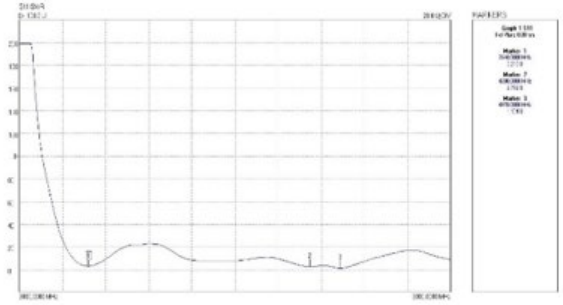
Fig. 7. is a validation of fabricated prototypes of the designed dual-band MPA's on an FR-4 substrate: (a) T-shaped antenna with DGS, (b) H-shaped antenna with coplanar parasitic patch elements, and (c) U-shaped antenna with coplanar parasitic patch elements, all fed using an SMA connector for experimental validation.



Fig(a)



Fig(b)



Fig(c)

Fig. 6. VSWR readings of Proposed Antennas (a) T-Shaped. (b) U-Shaped. (c) H-Shaped.

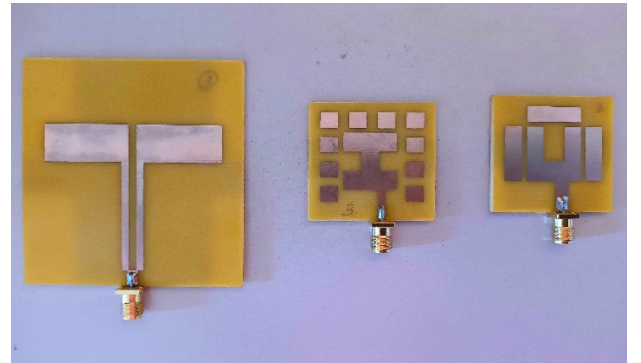


Fig. 7. Fabrication Image of the Proposed T-Shaped, U-Shaped and H-Shaped Antennas.

V. CONCLUSION

This work presented a comparative analysis of dual-band T-shaped, U-shaped, and H-shaped MPA's designed on a common FR-4 epoxy substrate for wireless applications operating around 5.25 GHz. Different performance enhancement techniques were adopted based on antenna geometry, where the T-shaped antenna employed a DGS technique to improve impedance bandwidth and radiation

efficiency, while the U-shaped and H-shaped antennas utilized coplanar parasitic patch elements for gain enhancement through near-field coupling. Simulation results confirm that the T-shaped antenna achieves the widest upper-band bandwidth (1.052 GHz) and the highest realized gain (6.15 dBi), whereas the U-shaped and H-shaped antennas provide stable dual-band performance with improved impedance matching and moderate gain of 5.2 dBi and 5.4 dBi respectively. All configurations exhibit VSWR values well below 2 at 5.41 GHz, ensuring effective power transfer and low reflection losses. Owing to their compact structure, low-cost implementation, and satisfactory impedance and radiation characteristics, the designed antennas are applicable for the WLAN, WiMAX, and other current wireless communication systems.

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